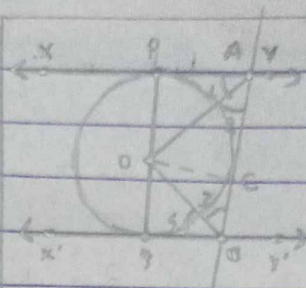


13. Given: A circle with centre O, $XY \parallel X'Y'$,
 $AB, XY, X'Y'$ are tangents



RTP: $\angle AOB = 90^\circ$

Proof: $AP = AC$ { The lengths of tangents
 $BC = BQ$ { drawn from external pts are equal

In $\triangle OAP$ and $\triangle OAC$

$OA = OA$ (common side)

$OP = OC$ (radii of same circle)

$AP = AC$ (proven above)

$\therefore \triangle OAP \cong \triangle OAC$ (SSS congruency)

$\Rightarrow \angle 1 = \angle 2$ (C.P.C.T) — (1)

Similarly, we can prove

$\angle 3 = \angle 4$ — (2)

W.K.T $XY \parallel X'Y'$, AB as transversal

$\angle PAB + \angle ABQ = 180^\circ$ (Co-interior Angs)

$(\angle 1 + \angle 2) + (\angle 3 + \angle 4) = 180^\circ$

$(\angle 2 + \angle 2) + (\angle 3 + \angle 3) = 180^\circ$ (from (1) & (2))

$2\angle 2 + 2\angle 3 = 180^\circ$

$2(\angle 2 + \angle 3) = 180^\circ$

$\angle 2 + \angle 3 = 90^\circ$ — (3)

In $\triangle AOB$

$\angle AOB = 180^\circ - (\angle 2 + \angle 3)$

$= 180^\circ - 90^\circ$ (from (3))

$\therefore \angle AOB = 90^\circ$

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8. (3) $7 - 4\sqrt{3}$

$$4 + 3 - 4\sqrt{3}$$

$$7 - 4\sqrt{3}$$

9. (4) 10cm

10. (2) $\frac{\sqrt{1+\cot^2 \theta}}{\cot \theta}$

$$\frac{1}{\sin \theta} \times \frac{\sin \theta}{\cos \theta}$$

11. Prob Given eqns- $\frac{2}{x} + \frac{3}{y} = 2$

$$\frac{1}{x} + \frac{1}{2y} = \frac{1}{3}$$

Let $\frac{1}{x} = a, \frac{1}{y} = b$

$$\Rightarrow 2a + 3b = 2 \quad \text{--- (1)}$$

$$a - \frac{1}{2}b = \frac{1}{3}$$

$$\Rightarrow 2a - b = \frac{2}{3}$$

*

$$2a + 3b = 2$$

$$-4b = \frac{2}{3} - 2$$

$$-4b = -\frac{4}{3}$$

$$b = \frac{1}{3}$$

$$\frac{1}{y} = \frac{1}{3}$$

$$y = 3$$

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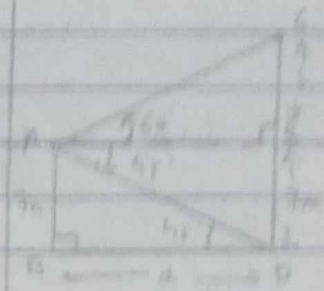
16(a) Given: Let $CE = h$

Let $BD = d = AE$

Height of building $AB = ED = 7\text{m}$

$\angle$ of elevation $= 60^\circ$

$\angle$ of depression $= 45^\circ$



To find: Height of tower $= CD = ?$

Soln: In ΔABD

$$\tan 45^\circ = \frac{7}{d}$$

$$1 = \frac{7}{d}$$

$$d = 7 \quad \text{--- (1)}$$

In ΔAEC

$$\tan 60^\circ = \frac{h}{d}$$

$$\sqrt{3} = \frac{h}{d}$$

$$h = d\sqrt{3} \quad \text{--- (2)}$$

Put (1) in (2)

$$h = 7\sqrt{3}\text{m}$$

$$= 7 \times (1.7)$$

$$= 11.9\text{m (approx)}$$

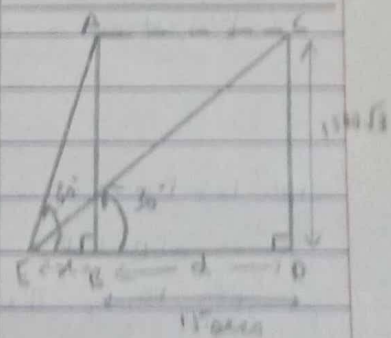
16(b) Given: Let $EB = x$

Let $BD = AC = d$

Height of jet fighter $= 1500\sqrt{3}\text{m}$

$\angle$ of elevation $= \theta_1 = 60^\circ$

$\theta_2 = 30^\circ$



To find: Speed of the jet $= ?$

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ALLEN PART TEST - 4

15. Total no. of cards = 52

$$n(S) = 52$$

(i) let A = a king of

$$P(E) = \frac{\text{No. of favourable outcomes w.r.t event}}{\text{Total no. of all possible outcomes}}$$

(i) let A = getting a king of red colour

$$n(A) = 2$$

$$P(\text{getting a king of red colour}) = \frac{2}{52} = \frac{1}{26}$$

(ii) let B = getting a face card

$$n(B) = 12$$

$$P(\text{getting a face card}) = \frac{12}{52} = \frac{3}{13}$$

(iii) let C = getting a red face card

$$n(C) = 6$$

$$P(\text{getting a red face card}) = \frac{6}{52} = \frac{3}{26}$$

(iv) let D = getting the jack of hearts

$$n(D) = 1$$

$$P(\text{getting the jack of hearts}) = \frac{1}{52}$$

(v) let E = getting a spade

$$n(E) = 13$$

$$P(\text{getting a spade}) = \frac{13}{52} = \frac{1}{4}$$

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Put $b = \frac{1}{3}$ in (1)

$$2a + 3\left(\frac{1}{3}\right) = 2$$

$$2a + 1 = 2$$

$$2a = 1$$

$$a = \frac{1}{2}$$

$$\frac{1}{x} = \frac{1}{2}$$

$$x = 2$$

$\therefore x = 2, y = 3$ are the required solutions

12. Given: $\triangle ABC$ and $\triangle DCB$ are rt $\triangle$ s, AC & BD intersect at P

To RTP: $AP \times PC = BP \times PD$

Proof: In $\triangle ABC$ & $\triangle DCB$

$$\angle PAB = \angle PDC = 90^\circ (\because \triangle ABC \text{ \& } \triangle DCB \text{ are rt } \triangle\text{s})$$

$$\angle APB = \angle DPC \text{ (Vertically opp. } \angle\text{s)}$$

$$\therefore \triangle APB \sim \triangle DPC \text{ (AA similarity)}$$

$$\Rightarrow \frac{AP}{DP} = \frac{PB}{PC} \text{ (CPST)}$$

$$\Rightarrow AP \times PC = BP \times PD$$

Hence Proved

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16-b) Soln: In $\triangle DAB$

$$\tan 60^\circ = \frac{1500\sqrt{3}}{x}$$

$$\sqrt{3} = \frac{1500\sqrt{3}}{x}$$

$$x = 1500$$

In $\triangle DCE$

$$\tan 30^\circ = \frac{1500\sqrt{3}}{d}$$

$$\frac{1}{\sqrt{3}} = \frac{1500\sqrt{3}}{d}$$

$$d = 1500\sqrt{3} \times \sqrt{3}$$

$$d = 4500$$

$\therefore$ Speed of the fighter jet = $\frac{d}{t}$

$$= \frac{4500}{15}$$

$$= 300 \text{ km/s}$$

$$= \frac{60}{300} \times 18$$

$\therefore$ Speed of the fighter jet = 1080 km/h

17. Let the speed of boat be x and the speed of still water be y

Relation 1: $x - y = 6$ — (1)

Relation 2: $x + y = 14$ — (2)

Solving (1) & (2)

$$\begin{array}{r} x - y = 6 \\ + \quad x + y = 14 \\ \hline \end{array}$$

$$2x = 20$$

$$x = 10$$

Put $x = 10$ in (2)

$$10 + y = 14$$

$$y = 4$$

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14. Consider, LMS

$$= (1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A)$$

$$= \left(1 + \frac{\cos A}{\sin A} - \frac{1}{\sin A}\right) \left(1 + \frac{\sin A}{\cos A} + \frac{1}{\cos A}\right)$$

$$= \left(\frac{\sin A + \cos A - 1}{\sin A}\right) \left(\frac{\cos A + \sin A + 1}{\cos A}\right)$$

$$= \frac{(\sin A + \cos A - 1)}{\sin A} \times \frac{(\sin A + \cos A + 1)}{\cos A}$$

$$= \frac{(\sin A + \cos A)^2 - 1^2}{\sin A \cos A}$$

$$= \frac{(\sin^2 A + \cos^2 A) + 2\sin A \cos A - 1}{\sin A \cos A}$$

$$= \frac{1 + 2\sin A \cos A - 1}{\sin A \cos A}$$

$$= \frac{2\sin A \cos A}{\sin A \cos A}$$

$$= 2$$

$$= \text{RMS}$$

17.(ii) (2) 10 km/h

17.(iii) (2) 4 km/h

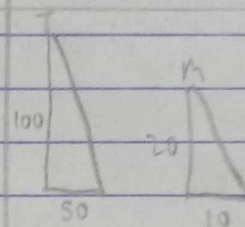
17.(iii) (2) Pair of linear equations

17.(iv) (2) downstream

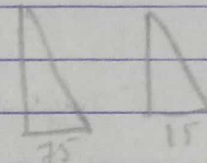
17.(v) (1) 7 km/h

$$\frac{10+4}{2} = 7$$

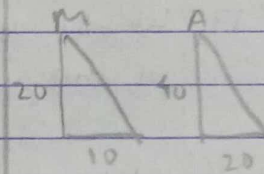
18.(i) (1) 100 m



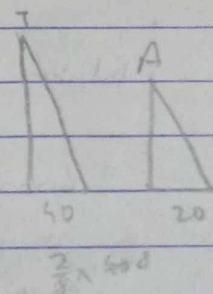
18.(ii) (3) 75 m



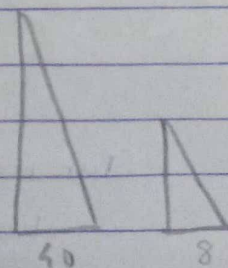
18.(iii) (4) 40 m



18.(iv) (2) 16 m



18.(v) (4) 8 m



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1. ~~2~~ (1) $\frac{1}{2}$

2. (2) 9 sq. units

$$x = y + 1 \Rightarrow 0$$

$$7x + 2y - 12 = 0$$

$$x + 2y + 0 = 0$$

$$5x + y - 11 = 0$$

3. (2) 2 cm

$$P + R = 5$$

$$P + 0 = 7$$

$$R + 0 = 6$$

$$P + 0 + R = 9$$

$$9 - 0 = 5$$

$$0 = 4$$

4. (1) 60

5. (4) no solution

$$0x + 2y = 9$$

$$0x + y = -7$$

6. (1) 2 cm

$$OA = 50$$

$$OC = 00$$

$$\frac{3x-1}{5x-3} = \frac{2x+1}{6x-5}$$

$$12x^2 - 15x - 6x + 5 = 10x^2 + 5x + 6x - 15$$

$$8x^2 - 20x + 8 = 0$$

$$8x^2 - 16x - 4x + 8 = 0$$

$$4x(x-2) - 4(x-2) = 0$$

$$x = 2, \quad \frac{4}{8} = \frac{1}{2}$$

7. (2) 1

$$\sin x + x - \cos^2 x = x$$

$$\sin x - \cos^2 x = 0$$